

2.0 HTML Table:Creating and using Tables,Basic Table Structure,Table Headers and Captions

2.0.0 HTML Table

An HTML table is a way to organize data in rows and columns. We create it with the `<table>` element in HTML. Tables mainly show tabular data like schedules, product lists, or reports.

Basic HTML Table Structure

An HTML table is created using the `<table>` tag. Inside the table, we use:

- [`<tr>`](#): Table Row
- [`<th>`](#): Table Header
- [`<td>`](#): Table Data

Basic Example:

```
index.html > html
1  <!-- index.html -->
2  <!DOCTYPE html>
3  <html lang="en">
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport"
7          content="width=device-width, initial-scale=1.0">
8      <title>HTML</title>
9  </head>
10 <body>
11     <table>
12         <tr>
13             <th>Firstname</th>
14             <th>Lastname</th>
15             <th>Age</th>
16         </tr>
17         <tr>
18             <td>Priya</td>
19             <td>Sharma</td>
20             <td>24</td>
21         </tr>
22         <tr>
23             <td>Arun</td>
24             <td>Singh</td>
25             <td>32</td>
26         </tr>
27         <tr>
28             <td>Sam</td>
29             <td>Watson</td>
30             <td>41</td>
31         </tr>
32     </table>
33 </body>
34 </html>
```

Output:

Firstname	Lastname	Age
Priya	Sharma	24
Arun	Singh	32
Sam	Watson	41

Tag	Description
<u><table></u>	Defines the structure for organizing data in rows and columns on a web page.
<u><tr></u>	Represents a row in an HTML table that contains individual cells.
<u><th></u>	Shows a table header cell, which usually holds titles or headings.
<u><td></u>	Represents a standard data cell that holds content or data.
<u><caption></u>	Provides a title or description for the entire table.
<u><thead></u>	Defines the header section of a table, often containing column labels.
<u><tbody></u>	Represents the main content area of a table, separating it from the header or footer.
<u><tfoot></u>	Specifies the footer section of a table that typically holds summaries or totals.
<u><col></u>	Defines attributes for table columns that can apply to multiple columns at once.
<u><colgroup></u>	Groups together a set of columns in a table so you can apply formatting or properties to all of them at once.

2.0.1 Basic Table Structure in HTML

HTML tables are not just a collection of rows and columns; they can be **logically grouped into sections** for better organization, accessibility, and styling. The three main grouping tags are:

- 1) **<thead>**: Groups header rows.
- 2) **<tbody>**: Groups body rows.
- 3) **<tfoot>**: Groups footer rows.

Example:

```
index.html > html > body > table
1  <!-- index.html -->
2  <!DOCTYPE html>
3  <html lang="en">
4  <head>
5      <meta charset="UTF-8">
6      <meta name="viewport"
7          content="width=device-width, initial-scale=1.0">
8      <title>HTML</title>
9  </head>
10 <body>
11     <table>
12         <thead>
13             <tr><th>Header 1</th><th>Header 2</th></tr>
14         </thead>
15         <tbody>
16             <tr><td>Data 1</td><td>Data 2</td></tr>
17         </tbody>
18         <tfoot>
19             <tr><td colspan="2">Footer</td></tr>
20         </tfoot>
21     </table>
22 </body>
23 </html>
```

Output

Header 1 Header 2	
Data 1	Data 2
Footer	

2.0.2 Table Headers and Captions

Headers and captions help organize and present data in tables.

The table heading is important because it labels the columns. The **<th>** (table header) element defines these headings.

Captions give a title or explanation for the table. The **<caption>** element is placed right after the opening table tag.

Employee Record		
Name	Age	City
Kelly Hu	21	Honolulu, Hawaii
Julia Roberts	23	Smyrna, Georgia

Syntax to Create Table's Header and Caption:

The following is the syntax to create a header and caption for an HTML table:

```
<table>
<caption>Description of table</caption>
<tr>
  <th>heading 1</th>
  <th>heading 2</th>
  <th>heading 3</th>
</tr>
</table>
```

2.1 HTML Table: Rowspan and Colspan, Table Attributes, Responsive Tables

2.1.0 HTML Table Colspan & Rowspan

HTML tables can have cells that span over multiple rows and/or columns.

NAME		

APRIL		

2022		
FIESTA		

HTML Table - Colspan

To make a cell span over multiple columns, use the colspan attribute:

Example:

```
index.html > html > body > table
1  <!DOCTYPE html>
2  <html>
3  <head>
4  <style>
5  table, th, td {
6      border: 1px solid black;
7      border-collapse: collapse;
8  }
9  </style>
10 </head>
11 <body>
12
13 <h2>Cell that spans two columns</h2>
14 <p>To make a cell span more than one column, use the colspan attribute.</p>
15
16 <table style="width:50%">
17     <tr>
18         <th colspan="2">Name</th>
19         <th>Age</th>
20     </tr>
21     <tr>
22         <td>Jill</td>
23         <td>Smith</td>
24         <td>43</td>
25     </tr>
26     <tr>
27         <td>Eve</td>
28         <td>Jackson</td>
29         <td>57</td>
30     </tr>
31 </table>
32 </body>
33 </html>
```

Output:

Cell that spans two columns

To make a cell span more than one column, use the colspan attribute.

Name		Age
Jill	Smith	43
Eve	Jackson	57

HTML Table - Rowspan

To make a cell span over multiple rows, use the rowspan attribute:

Example:

```
index.html > html > body > table
1  <!DOCTYPE html>
2  <html>
3  <head>
4  <style>
5  table, th, td {
6    border: 1px solid black;
7    border-collapse: collapse;
8  }
9  </style>
10 </head>
11 <body>
12
13 <h2>Cell that spans two rows</h2>
14 <p>To make a cell span more than one row, use the rowspan attribute.</p>
15
16 <table style="width:50%">
17   <tr>
18     <th>Name</th>
19     <td>Jill</td>
20   </tr>
21   <tr>
22     <th rowspan="2">Phone</th>
23     <td>555-1234</td>
24   </tr>
25   <tr>
26     <td>555-8745</td>
27   </tr>
28 </table>
29 </body>
30 </html>
31
```

Output:

Cell that spans two rows

To make a cell span more than one row, use the rowspan attribute.

Name	Jill
Phone	555-1234
	555-8745

2.1.1 HTML Table Attributes

HTML tables can be customized using attributes to control appearance, spacing, and alignment. Many of these attributes are outdated in HTML5, but they still help in understanding older code. For modern projects, CSS is the preferred option.

Most common attributes are:

- **border:** The **HTML <table> border Attribute** is used to specify the border of a table. It sets the border around the table cells. This attribute defines the visual presentation of the table by setting the thickness of the borders. A higher value results in a thicker border. Alternatively, setting the border attribute to "0" removes the borders entirely.

Syntax:

<table border="value">

The value represents the thickness of the border in pixels.

border="1": This means the table will have a border of 1 pixel.

border="0": This will remove the borders from the table, leaving it borderless.

```

1 <html>
2 <head>
3 <style>
4 table, th, td {
5     border: 1px solid black;
6     border-collapse: collapse;
7 }
8 </style>
9 </head>
10 <body>
11     <h1>This is Border Attribute</h1>
12     <h2>HTML table border Attribute</h2>
13     <table border="1">
14         <caption>
15             Author Details
16         </caption>
17
18         <tr>
19             <th>NAME</th>
20             <th>AGE</th>
21             <th>BRANCH</th>
22         </tr>
23         <tr>
24             <td>Shalini</td>
25             <td>22</td>
26             <td>CSE</td>
27         </tr>
28         <tr>
29             <td>RAM</td>
30             <td>21</td>
31             <td>ECE</td>
32         </tr>
33     </table>
34 </body>
35 </html>

```

Output:

This is Border Attribute		
HTML table border Attribute		
Author Details		
NAME	AGE	BRANCH
Shalini	22	CSE
RAM	21	ECE

- **Cellpadding:** The HTML `<table>` **cell padding Attribute** is used to specify the space between the cell content and cell wall. The cellpadding attribute is set in terms of pixels. By default, the padding is set to 0.

Syntax: `<table cellpadding="pixels">`

Note: The `<table>` cellpadding Attribute is not supported by [HTML5](#).

```
index.html x
index.html > html > body > h1
1 <html>
2 <head>
3   <title>
4     HTML table cellpadding Attribute
5   </title>
6 </head>
7
8 <body>
9   <h1>HTML TABLE Attribute</h1>
10
11   <h2>
12     HTML table cellpadding Attribute
13   </h2>
14
15   <table border="1" cellpadding="15">
16     <caption>Author Details</caption>
17
18     <tr>
19       <th>NAME</th>
20       <th>AGE</th>
21       <th>BRANCH</th>
22     </tr>
23     <tr>
24       <td>BITTU</td>
25       <td>22</td>
26       <td>CSE</td>
27     </tr>
28     <tr>
29       <td>RAM</td>
30       <td>21</td>
31       <td>ECE</td>
32     </tr>
33   </table>
34 </body></html>
```

Output:

HTML TABLE Attribute		
HTML table cellpadding Attribute		
Author Details		
NAME	AGE	BRANCH
BITTU	22	CSE
RAM	21	ECE

- **Cellspacing:** The cellspacing attribute in HTML is used to define the space between cells in a table. It controls the amount of space, in pixels, between the table cells. This spacing can make the table's content more readable by adding extra room between cells.

Syntax: <table cellspacing="pixels">

```
index.html > html > head > title
1  <html>
2  /
3
4  <body>
5
6      <h1 style="color: green;">
7          HTML table Attribute
8      </h1>
9
10     <h2>HTML table cellspacing Attribute</h2>
11
12     <table border="1" cellspacing="15">
13         <caption>
14             Author Details
15         </caption>
16
17         <tr>
18             <th>NAME</th>
19             <th>AGE</th>
20             <th>BRANCH</th>
21         </tr>
22         <tr>
23             <td>BITTU</td>
24             <td>22</td>
25             <td>CSE</td>
26         </tr>
27         <tr>
28             <td>RAM</td>
29             <td>21</td>
30             <td>ECE</td>
31         </tr>
32     </table>
33 </body></html>
34
35
```

Output:

HTML table Attribute		
HTML table cellpadding Attribute		
Author Details		
NAME	AGE	BRANCH
BITTU	22	CSE
RAM	21	ECE

- **Align:** The **HTML <table> align Attribute** specify the table's alignment but CSS properties like margin and text-align are preferred for table alignment. To align content within table rows or cells, use the align attribute within [<tr>](#) and [<td>](#) tag or apply CSS styles

Syntax:

<table align="left | right | center">

Table Alignment Attributes	
Attribute	Description
left	Aligns the table to the left side of the page. This is the default setting.
right	Aligns the table to the right side of the page.
center	Aligns the table in the horizontal center of the page.

Example:

```
index.html X
index.html > html
1  <html><head>
2      <title>
3          HTML table align Attribute
4      </title>
5  </head>
6
7  <body>
8
9      <h2>HTML table align Attribute</h2>
10     <table align="left" border="1">
11         <caption>
12             Author Details
13         </caption>
14         <tr>
15             <th>NAME</th>
16             <th>AGE</th>
17             <th>BRANCH</th>
18         </tr>
19         <tr>
20             <td>BITTU</td>
21             <td>22</td>
22             <td>CSE</td>
23         </tr>
24         <tr>
25             <td>RAM</td>
26             <td>21</td>
27             <td>ECE</td>
28         </tr>
29     </table>
30 </body>
31 </html>
32
```

Output:

HTML table align Attribute		
Author Details		
NAME	AGE	BRANCH
BITTU	22	CSE
RAM	21	ECE

2.2 HTML Table: Nested Tables, Accessibility in Tables, Practical

2.2.0 Nested Tables

In HTML, tables are created using the `<table>` tag. To nest a table, put one table inside another. The outer table is your main table, while the inner table is the one you are nesting inside.

To create nested Table:

- First, you create the outer table using the `<table>` tag.
- Then, within one of the `<td>` tags (which are used to define individual cells in the table), you insert another `<table>` tag to create the inner table.

Example:

```
index.html x
index.html > html > body > table > tr > td > table > tr
1  <html><head>
5  </head>
6  <body>
7      <h2>About Table</h2>
8      <p><b>Nested tables</b></p>
9
10     <table border="2">
11         <tr>
12             <td>Main Table row 1 column 1</td>
13             <td>Main Table column 2
14                 <table border="2">
15                     <tr>
16                         <td>Inner Table row 1 column 1</td>
17                         <td>Inner Table row 1 column 2</td>
18                     </tr>
19                     <tr>
20                         <td>Inner Table row 2 column 1</td>
21                         <td>Inner Table row 2 column 2</td>
22                     </tr>
23                     <tr>
24                         <td>Inner Table row 3 column 1</td>
25                         <td>Inner Table row 3 column 2</td>
26                     </tr>
27                 </table>
28             </td>
29         </tr>
30         <tr>
31             <td>Main Table row 2 column 1</td>
32             <td>Main Table row 2 column 2</td>
33         </tr>
34     </table>
35 </body>
36 </html>
```

Output:

Nested tables

Main Table row 1 column 1	Main Table column 2	
	Inner Table row 1 column 1	Inner Table row 1 column 2
	Inner Table row 2 column 1	Inner Table row 2 column 2
	Inner Table row 3 column 1	Inner Table row 3 column 2
Main Table row 2 column 1	Main Table row 2 column 2	

2.2.1 HTML Table Accessibility

Tables help organize data visually, but **screen readers require additional information** to interpret them correctly. Without proper markup, users with visual impairments cannot understand the relationship between rows and columns as easily as sighted users. This is why **semantic HTML elements** are crucial.

Main Accessibility Features:

1. **Use Semantic Tags (<thead>, <tbody>, <tfoot>)**

These tags provide structure and meaning to your table:

- a) **<thead>**: Groups the header rows of the table.
- b) **<tbody>**: Contains the main body rows (data rows).
- c) **<tfoot>**: Groups the footer rows, often used for totals or summaries.

Using these tags helps assistive technologies and search engines understand the role of each section. It also improves styling and layout control with CSS.

2. **Add scope Attribute for Headers**

The scope attribute clarifies whether a header cell applies to a row, a column, or a group:

- a) **scope="col"** → Column header
- b) **scope="row"** → Row header
- c) **scope="colgroup"** → Multiple columns
- d) **scope="rowgroup"** → Multiple rows

This ensures that screen readers can associate the correct headers with data cells, improving accessibility for visually impaired users.

3. **Use <caption> for Description**

The <caption> tag provides a brief description of the table's purpose and content.

- Placed immediately after the <table> tag.
- Helps users (including those using screen readers) understand the context without reading all the data.

Example:

```
<table>
  <caption>Student Attendance Report</caption>
</table>
```

4. Add aria-describedby for Screen Readers

The `aria-describedby` attribute can link the table to additional descriptive text (like a summary or explanation) outside the table.

Example:

```
<div id="table-info">This table shows student attendance for the month
of July.</div>
<table aria-describedby="table-info">
  ...
</table>
```

5. Avoid Using Tables for Layout Purposes

Tables should **only** be used for displaying tabular data—not for page layout.

- Using tables for layout creates confusion for screen readers and breaks semantic structure.
- For layouts, use **CSS grid or flexbox** instead.

2.2.2 Practical Questions

Creating and Using Tables

1. Create a simple HTML table to display the following student data:
Name, Roll No, Subject, Marks for at least **5 students**.
2. Add a **caption** to the table describing the data.
3. Apply basic table structure with `<thead>`, `<tbody>`, and `<tfoot>` tags.

```
index.html x
index.html > html > body > table
1 <html><head>
2   <title>
3     HTML table
4   </title>
5 </head>
6 <body>
7   <table border="">
8     <caption>Student Marks Data</caption>
9     <thead>
10      <tr>
11        <th scope="col">Name</th>
12        <th scope="col">Roll No</th>
13        <th scope="col">Subject</th>
14        <th scope="col">Marks</th>
15      </tr>
16    </thead>
17    <tbody>
18      <tr><td>Amit</td><td>101</td><td>Math</td><td>90</td></tr>
19      <tr><td>Riya</td><td>102</td><td>Science</td><td>88</td></tr>
20      <tr><td>Karan</td><td>103</td><td>English</td><td>75</td></tr>
21      <tr><td>Pooja</td><td>104</td><td>Math</td><td>82</td></tr>
22      <tr><td>Rahul</td><td>105</td><td>Science</td><td>95</td></tr>
23    </tbody>
24    <tfoot>
25      <tr>
26        <td colspan="4">Total Students: 5</td>
27      </tr>
28    </tfoot>
29  </table>
30
31 </body>
32 </html>
33
```

Output:

Student Marks Data			
Name	Roll No	Subject	Marks
Amit	101	Math	90
Riya	102	Science	88
Karan	103	English	75
Pooja	104	Math	82
Rahul	105	Science	95
Total Students: 5			

Rowspan and Colspan

Create a timetable for your class using HTML tables:

1. Use rowspan for subjects that cover multiple time slots.
2. Use colspan for breaks or combined sessions.

```
index.html x
index.html > html > body > table > tbody > tr > td

 2 <html>
 3 <head>
 4
 5
 6 </head>
 7 <body>
 8     <table border="1" align="center">
 9         <caption>Class Timetable</caption>
10         <thead>
11             <tr>
12                 <th>Day</th>
13                 <th>8am-9am</th>
14                 <th>9am-10am</th>
15                 <th>10am-11am</th>
16                 <th>12pm-1pm</th>
17             </tr>
18         </thead>
19         <tbody>
20             <tr>
21                 <td>Mon</td>
22                 <td rowspan="2">Math</td>
23                 <td>Science</td>
24                 <td colspan="2">Lab</td>
25             </tr>
26             <tr>
27                 <td>Tue</td>
28                 <td>English</td>
29                 <td>Break</td>
30                 <td>History</td>
31             </tr>
32         </tbody>
33     </table>
34 </body>
35 </html>
36
```

Output:

Class Timetable				
Day	8am-9am	9am-10am	10am-11am	12pm-1pm
Mon	Math	Science	Lab	
Tue		English	Break	History

Nested Tables

Create a table where one of the cells contains **another table** (nested table) showing additional details.

```
index.html x
index.html > html > head > title
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Order List</title>
5
6 </head>
7 <body>
8   <table border="1" align="center">
9     <caption>Order Details</caption>
10    <tr>
11      <th>Order ID</th>
12      <th>Customer</th>
13      <th>Items</th>
14    </tr>
15    <tr>
16      <td>101</td>
17      <td>Amit</td>
18      <td>
19        <table>
20          <tr><th>Item</th><th>Qty</th></tr>
21          <tr><td>Laptop</td><td>1</td></tr>
22          <tr><td>Mouse</td><td>2</td></tr>
23        </table>
24      </td>
25    </tr>
26  </table>
27 </body>
28 </html>
29
30
```

Output:

Order Details								
Order ID	Customer	Items						
101	Amit	<table><tr><th>Item</th><th>Qty</th></tr><tr><td>Laptop</td><td>1</td></tr><tr><td>Mouse</td><td>2</td></tr></table>	Item	Qty	Laptop	1	Mouse	2
Item	Qty							
Laptop	1							
Mouse	2							